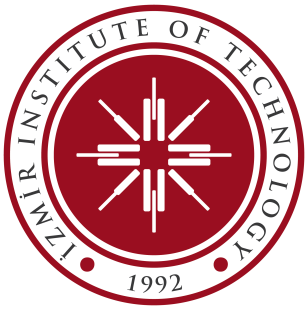


İzmir Institute of Technology

Faculty of Engineering

Department of Electrical and Electronics Engineering

EE 315 Electronics Laboratory Prelab Sheet

Experiment 2

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Preliminary Work

Problem 1. Full-Wave Bridge Rectifier

a) Diode Conduction

In the full-wave bridge rectifier, the conduction alternates based on the polarity of the transformer secondary voltage v_S:

- **Positive Half-Cycle ($v_S > 0$):** Current flows from the positive upper terminal through D_1, across the load R_L, and returns through D_3 to the lower terminal. Thus, D_1 **and** D_3 **conduct**.
- **Negative Half-Cycle ($v_S < 0$):** The lower terminal becomes positive. Current flows through D_2, across the load R_L in the same direction, and returns through D_4 to the upper terminal. Thus, D_2 **and** D_4 **conduct**.

b) Output Frequency

Because the full-wave rectifier inverts the negative half-cycles to the positive side, the fundamental frequency of the output ripple v_L is twice the input frequency:

$$f_L = 2f_S = 2(50\text{ Hz}) = \mathbf{100}\text{ Hz}$$

c) Total Average Power Dissipation in Diodes

Given $V_{SP} = 9$ V and $V_{\text{ON}} = 0.6$ V. Two diodes are in series during conduction, so the peak load voltage is:

$$V_{LP} = V_{SP} - 2V_{\text{ON}} = 9\text{ V} - 1.2\text{ V} = 7.8\text{ V}$$

The average (DC) load voltage is:

$$V_{LDC} = \frac{2V_{LP}}{\pi} = \frac{2(7.8\text{ V})}{\pi} \approx 4.966\text{ V}$$

The average DC current through the load $R_L = 1.2\,\text{k}\Omega$ is:

$$I_{LDC} = \frac{V_{LDC}}{R_L} = \frac{4.966\text{ V}}{1200\,\Omega} \approx 4.138\text{ mA}$$

Since the total load current always passes through two conducting diodes, the total average power dissipated by the diode bridge is:

$$P_{D,\text{total}} = I_{LDC} \times (2V_{\text{ON}}) = 4.138\text{ mA} \times 1.2\text{ V} \approx \mathbf{4.966}\text{ mW}$$

Problem 3. Full-Wave Bridge Rectifier with $C_1 = 22\,\mu$F Filter

Because the time constant $\tau = R_L C_1 = 330\,\Omega \times 22\,\mu\text{F} = 7.26$ ms is shorter than the 10 ms ripple period, the capacitor discharges deeply. The standard triangular ripple approximation ($V_{rplPP} = \frac{V_{LDC}}{fR_LC}$) fails under these large-ripple conditions. We must use the exact exponential discharge method.

a) Load Regulation
The peak load voltage is $V_{LP} = 9\text{ V} - 2(0.6\text{ V}) = 7.8$ V. At no-load ($R_L \to \infty$), the capacitor does not discharge, so $V_{NL} = V_{LP} = \mathbf{7.8}$ V.

At full load ($R_L = 330\,\Omega$), the capacitor voltage decays exponentially from the peak at $t = 0$:

$$v_C(t) = V_{LP} e^{-t/\tau} = 7.8 e^{-t/7.26\text{ ms}}$$

The rectifier diodes turn back on when this decaying voltage intersects the rising edge of the next rectified half-sine wave, given by $v_{\text{rect}}(t) = -9\cos(100\pi t) - 1.2$. Setting them equal:

$$7.8 e^{-t/7.26\text{ ms}} = -9\cos(100\pi t) - 1.2$$

Solving this transcendental equation numerically yields an intersection time of $t \approx 6.62$ ms. Substituting this back into the discharge equation gives the minimum trough voltage:

$$V_{\text{min}} = 7.8 e^{-6.62/7.26} \approx 3.13\text{ V}$$

The average DC voltage is approximately the midpoint of the peak and trough:

$$V_{FL} = V_{LDC} \approx \frac{V_{LP} + V_{\text{min}}}{2} = \frac{7.8 + 3.13}{2} = 5.465\text{ V}$$

$$\%\text{ Load Regulation} = \frac{V_{NL} - V_{FL}}{V_{NL}} \times 100\% = \frac{7.8 - 5.465}{7.8} \times 100\% = \mathbf{29.93}\%$$

b) Ripple Factor
The peak-to-peak ripple voltage is $V_{rplPP} = 7.8\text{ V} - 3.13\text{ V} = 4.67$ V. Using the definition from the manual:

$$\text{Ripple Factor} = \frac{V_{rplPP}}{2\sqrt{3}V_{LDC}} = \frac{4.67}{2\sqrt{3}(5.465)} = \frac{4.67}{18.93} = \mathbf{24.67}\%$$

(Note: As R_L increases to 1.2 kΩ and 10 kΩ, the time constant τ becomes very large, and the standard linear approximation becomes accurate again, yielding $\approx 10.9\%$ and $\approx 1.3\%$ respectively.)

c) Effect of Load Resistance
As the load resistance R_L increases, the time constant τ increases. The capacitor discharges at a much slower exponential rate between cycles, shifting the intersection point closer to the peak. This results in a smaller peak-to-peak ripple voltage and a higher, smoother DC output.

Problem 4. Capacitive Filtering

a) Load Regulation with Added Capacitor

Adding a second $22\,\mu$F capacitor in parallel makes $C_{tot} = 44\,\mu$F. The no-load voltage remains $V_{NL} = 7.8$ V. The new time constant is $\tau = 330\,\Omega \times 44\,\mu\text{F} = 14.52$ ms. Setting the new exponential discharge equal to the rectified wave:

$$7.8e^{-t/14.52\text{ ms}} = -9\cos(100\pi t) - 1.2$$

Solving numerically gives $t \approx 7.28$ ms. The new trough voltage is:

$$V_{\min} = 7.8e^{-7.28/14.52} \approx 4.72 \text{ V}$$

$$V_{FL} = V_{LDC} = \frac{7.8 + 4.72}{2} = 6.26 \text{ V}$$

$$\%\text{ Load Regulation} = \frac{7.8 - 6.26}{7.8} \times 100\% = \mathbf{19.74}\%$$

b) Effect of Increasing Filter Capacitance

Increasing the filter capacitance directly increases the *RC* time constant. This slows the voltage decay rate during the load discharge phase, preventing the voltage from dipping as low before the next cycle. The result is a significantly reduced ripple voltage (4.67 V down to 3.08 V) and an improved load regulation percentage.

5. LTspice Simulation

0.1 Full-Wave Bridge Rectifier

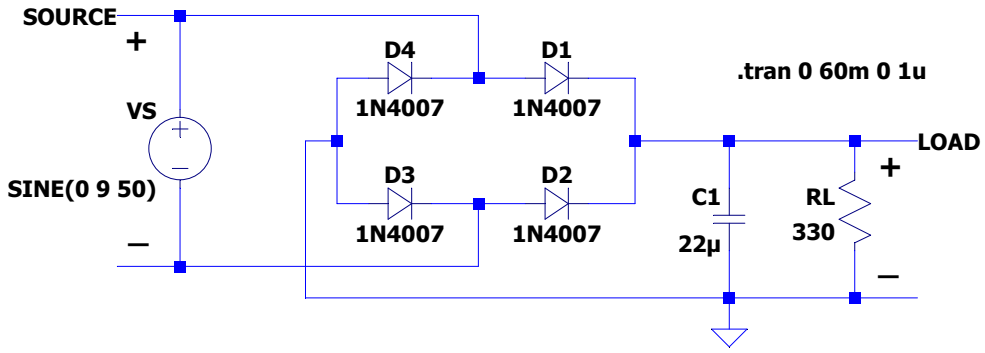


Figure 1: Schematic of full-wave bridge rectifier circuit

$V_{L,\text{peak}}$	f_L	**Number of conducting diodes**
7.59 V	100 Hz	2

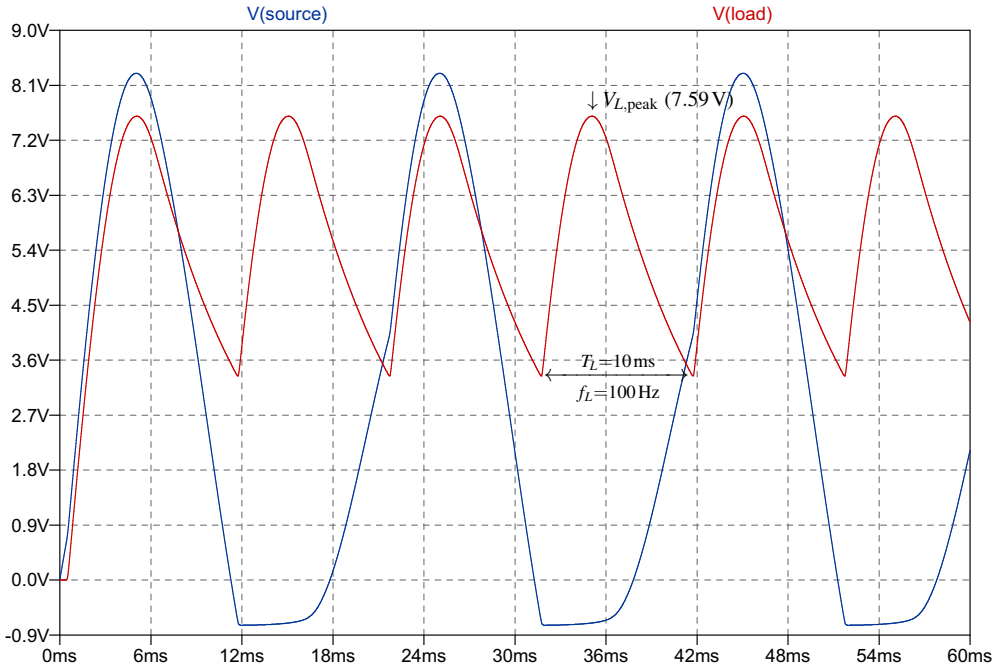


Figure 2: Input and output voltage waveforms of the full-wave bridge rectifier circuit

0.2 Zener Diode Regulator

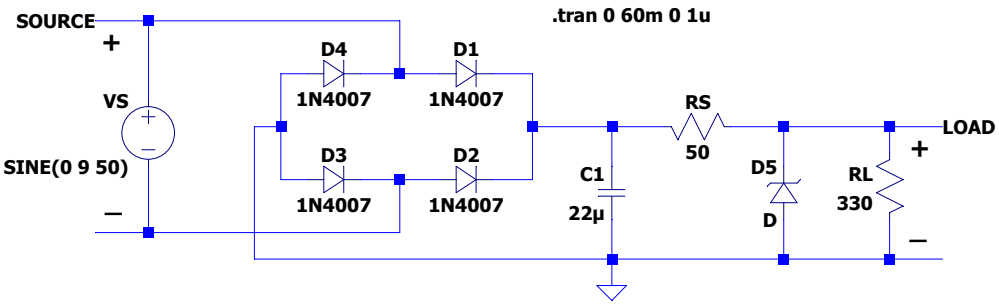


Figure 3: Schematic of Zener diode regulator circuit

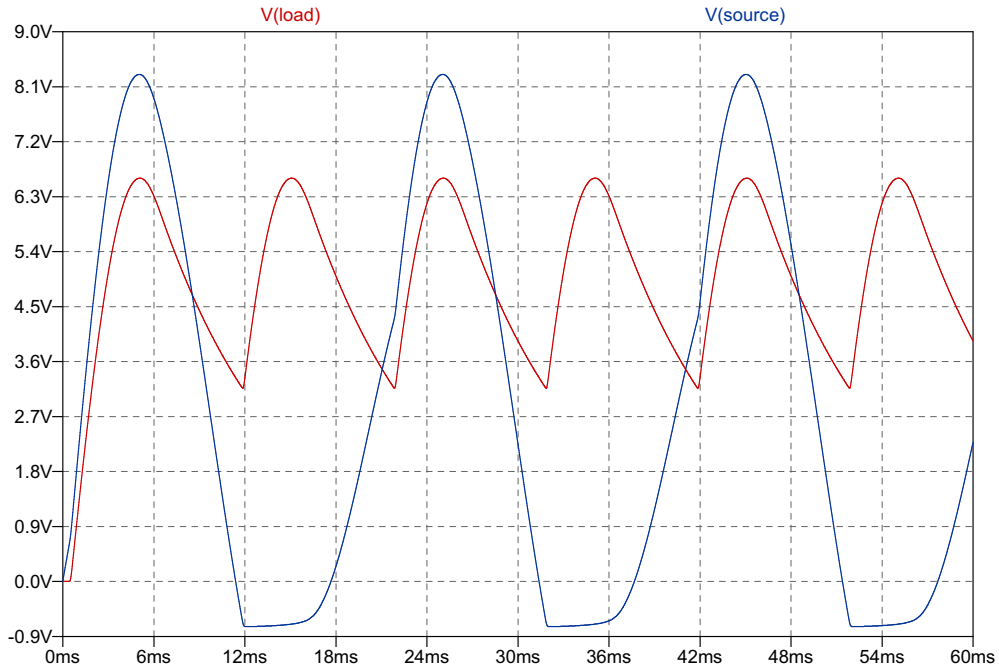


Figure 4: Input and output voltage waveforms of the Zener diode regulator circuit

$V_{o,DC}$	I_Z
TBC	TBC